Platform Engineering 101: Building Internal Developer Platforms

Maarten
Red Hat

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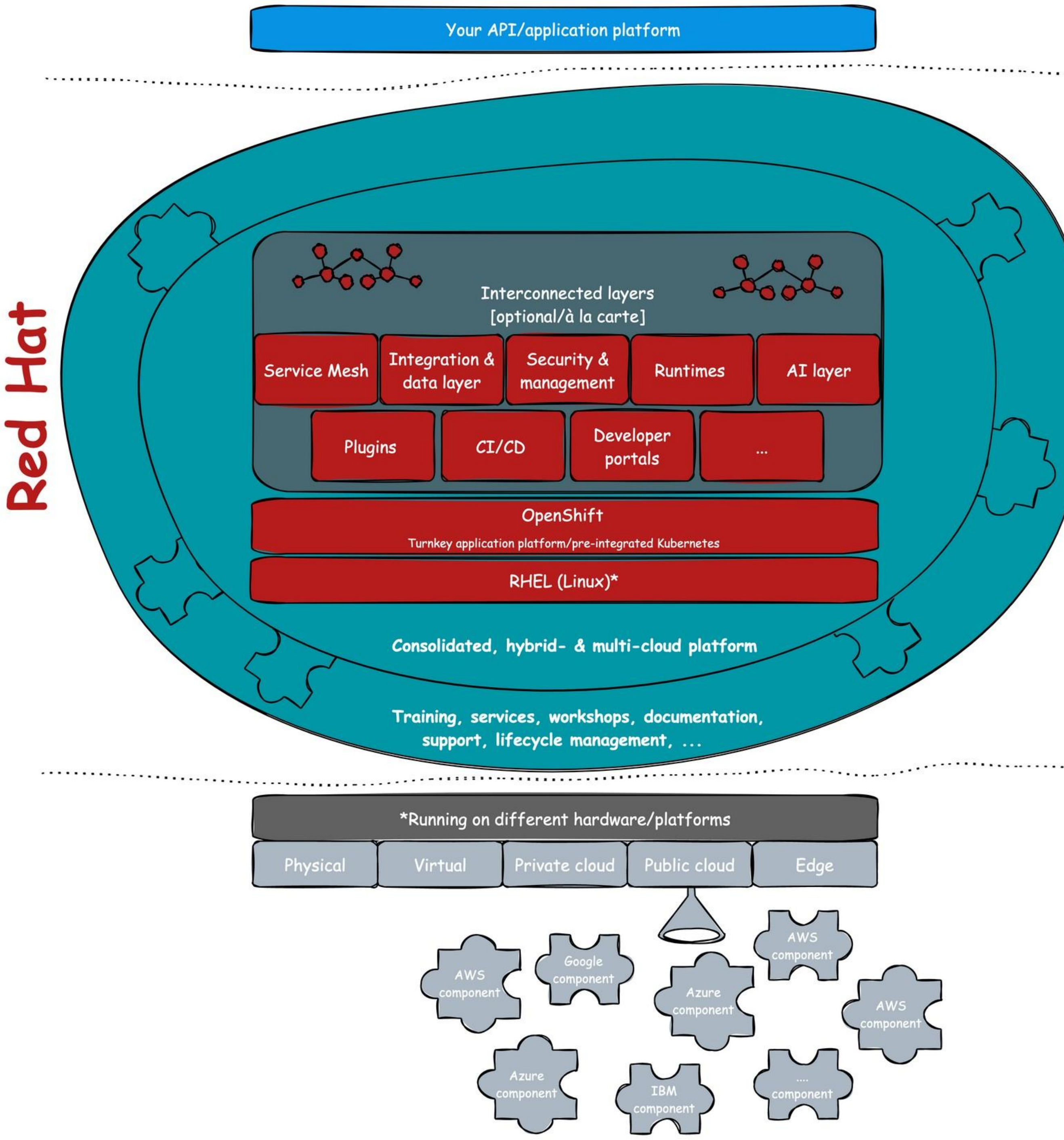
maarten-vandeperre.be

Source: AI

Maarten Vandeperre

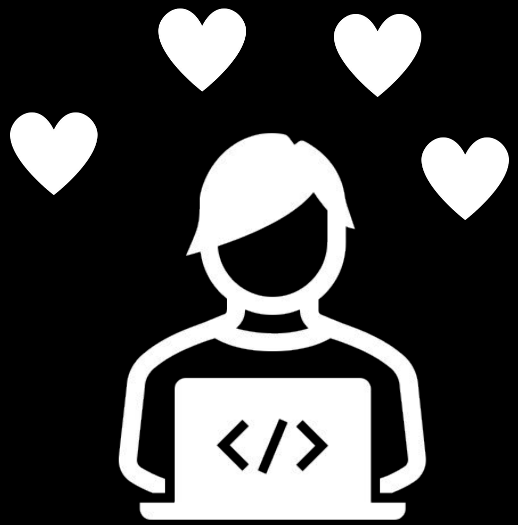


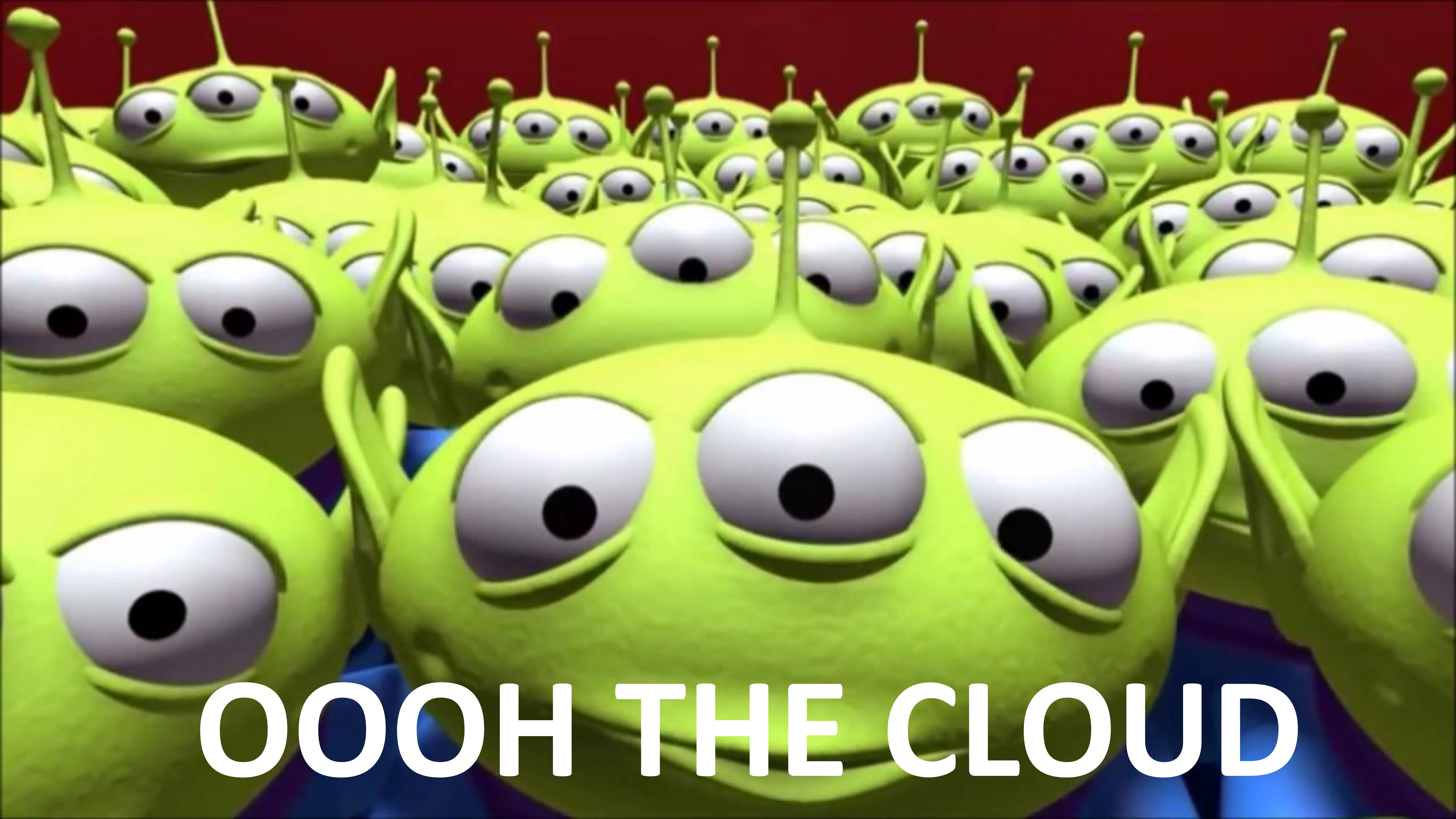
Frustration matrix

Frustrations between "silos"

FRUSTRATED WITH		Data scientist	App developer API	App developer AI embedding	Business (owner)	Platform team	Security officer
I need (production) data, I need big data sets and I want data changes and extra data now	Data scientist		- Data (contract) changes dev is too slow (for rapid prototyping) - Extra data (fields) dev is too slow - API agile/SCRUM dev is too slow, too mature - I don't need APIs, I need all the data - "It works on my machine" - Messy or inaccessible data		- I just want to have my tooling available, don't care about an enterprise-grade, centralized solution - Collaboration barriers - unclear or shifting requirements	- We don't need the same platform/processes as development teams - We just want access to our data - It takes too long to get access to new databases or other technologies or it's too slow	- Too much time is lost with implementing all these security requirements
I need a rock-solid platform, modern tooling, CI/CD-tooling and scalable infrastructure	App developer API	- Your requests are not matching our API development process - Your requested data contracts are conflicting with the ones from our (application) APIs			- Requirements are not clear - We're often delivering too slow according to you - Requirements are often changing - Often a fight to get new platforms/technologies paid	- Platform is not flexible enough - Platform is not intuitive enough - Platform is lacking features - No proper CI/CD infrastructure - We don't always get our extra technology (e.g., new database) - only one environment/platform	- Too much time is lost with implementing all these security requirements
I need stable models, seamless inference APIs, and rapid model iteration	App developer AI embedding	- Model format & compatibility - Dependency & version hell	- App platform is not flexible enough to extend it with our functionality (which are APIs too)		- No real business needs to embed AI, just need to do AI because our competitors are doing this too - AI's impact is unclear, no ROI		
I want to make money and need to have features delivered fast, without downtime	Business (owner)	- No or limited focus on data privacy/data security	- Features from idea to production release => too slow - Release cycle not aligned with business - Often issues with deployments/releases - "It works on my machine"			- Platform faces downtime every now and then - Platform should improve dev efficiency more - Platform instability and untested changes lead to downtime	
I need standardized tooling, platform requirements from developers and SLAs from business	Platform team	- Shadow IT - Working around the platform	- Shadow IT - Working around the platform	- Shadow IT - Working around the platform	- Money discussions => off the shelf versus DIY - Unclear requirements, SLAs, ...		- Lack of clear security guidelines for (AI) deployments
I want to have the platform as secure as possible to avoid downtime, data leakage and offensive responses	Security officer	- No or limited focus on API and/or container security - No auditing on third party AI components - Data silos & visibility gap	- No or limited focus on API and/or container security - No encryption/decryption on all endpoints - No zero-trust networks - Shadow IT & unmanaged assets	- No or limited focus on API and/or container security - No auditing on third party AI components - Shadow AI	- Compliance pressure	- AI without insights on data is a problem - Tool sprawl and integration gaps - Extra tooling: evolving threat landscape	

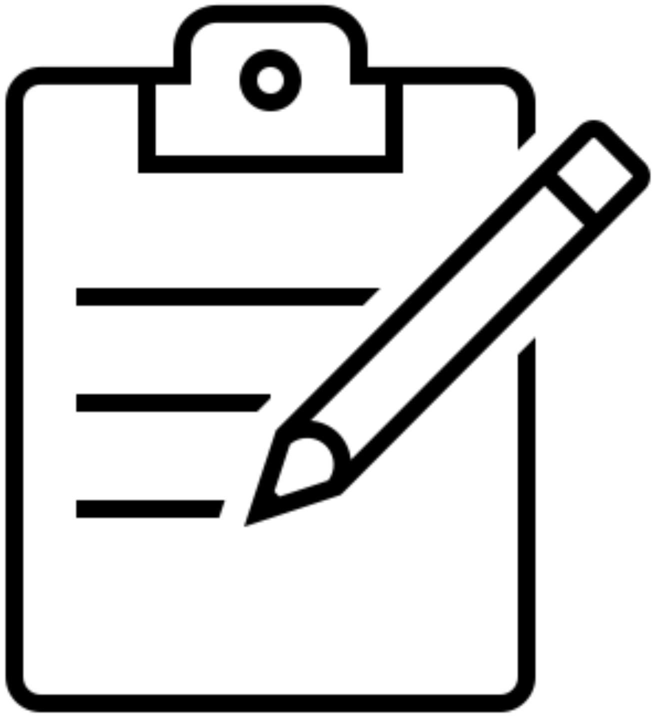
Let’s get started




OOOH THE CLOUD

Agenda

- How Did We Get Here?
- Developer Struggles
- Platform Engineering
- Internal Developer Platform (IDP)
- Internal Developer Portal
- Benefits for Developers
- Tools to Help
- Demo (25-30 min!!)
- Summary & Resources



What Happened?!

Shift Left: Increasing cognitive load

It’s not just app development anymore

Dev

Ops

DevOps

FinOps

MLOps

GitOps

DataOps

DevSecOps

BizDevOps

BizDevSecOps

BizDevSecFinMLOps

Organizational Impact & Cognitive Load

Traditional App Delivery

Dev

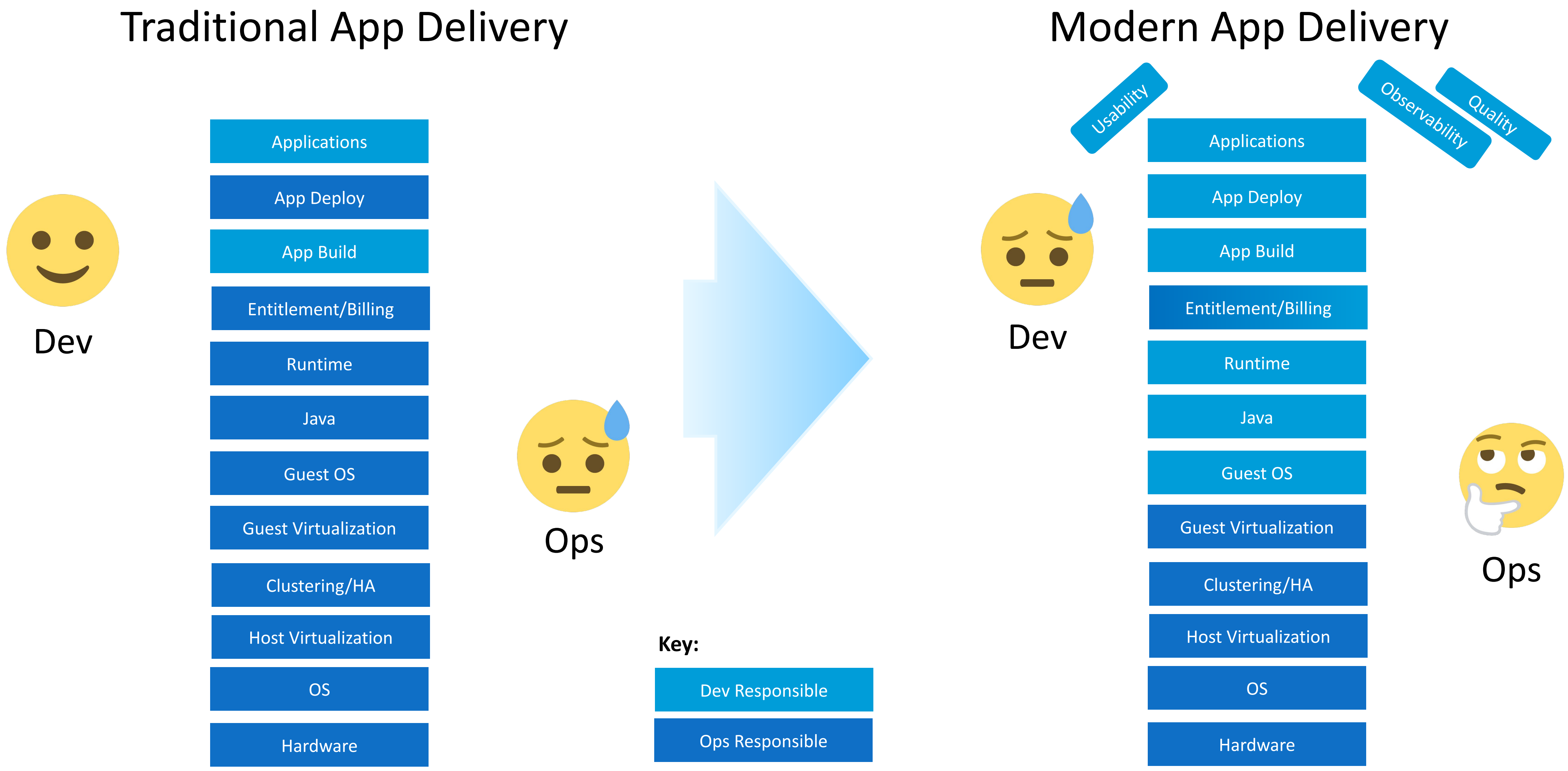
- Applications
- App Deploy
- App Build
- Entitlement/Billing
- Runtime
- Java
- Guest OS
- Guest Virtualization
- Clustering/HA
- Host Virtualization
- OS
- Hardware

Ops

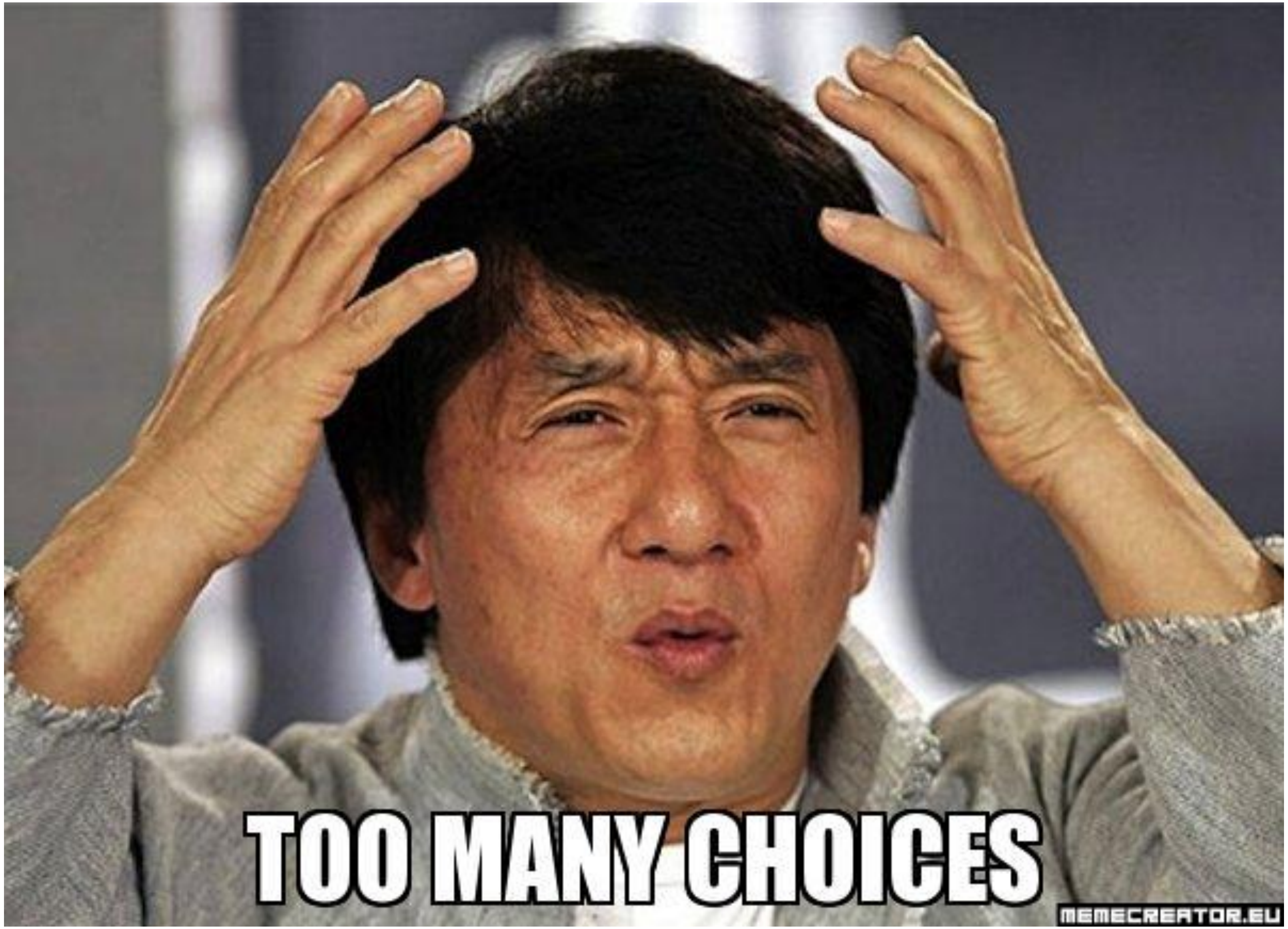
Key:

- Dev Responsible
- Ops Responsible

Organizational Impact & Cognitive Load



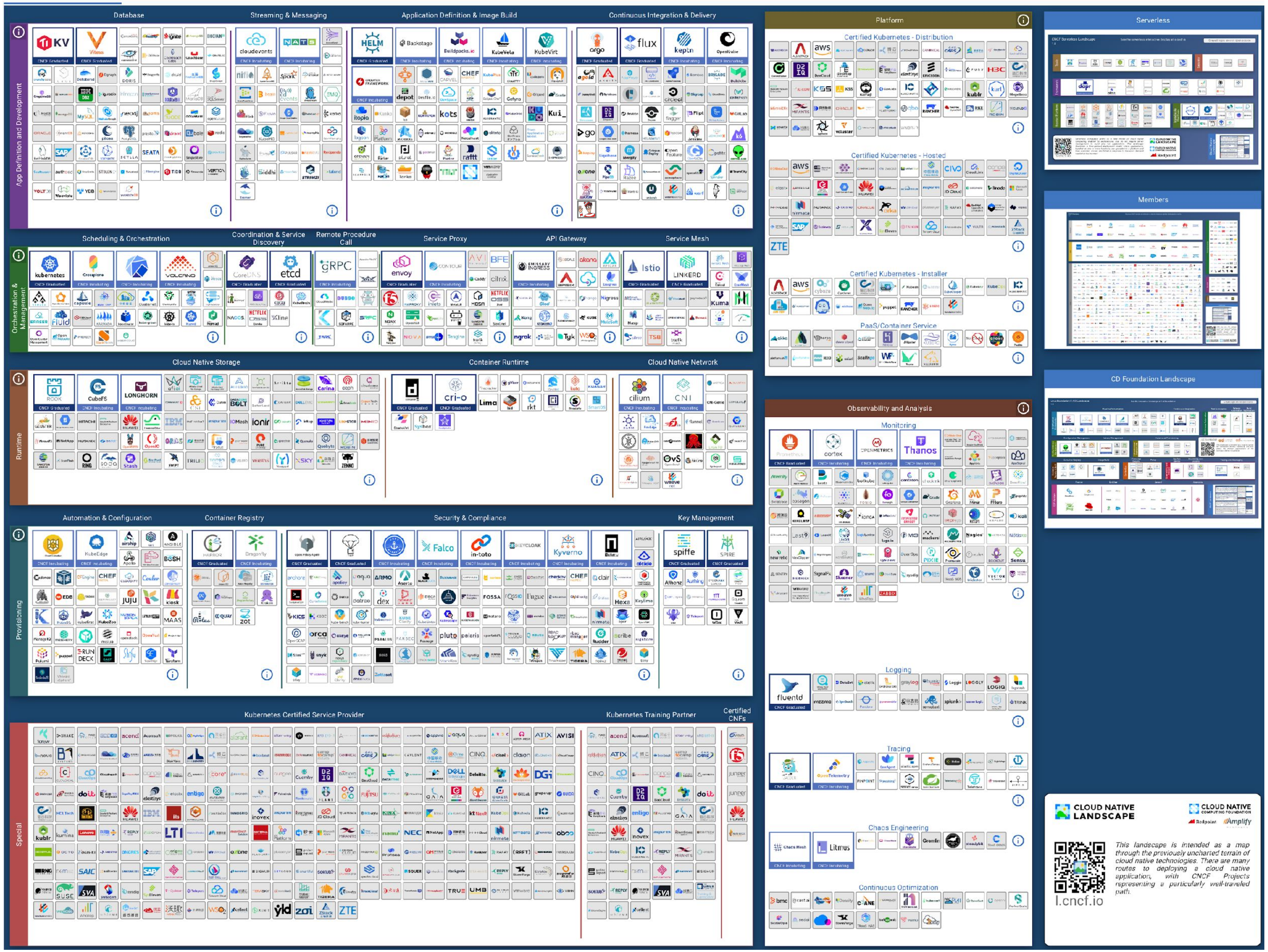
Tools:
Too much
choice



The Cloud-Native Grocery Store

Where's Waldo?

or Wally?



Source: https://landscape.cncf.io/

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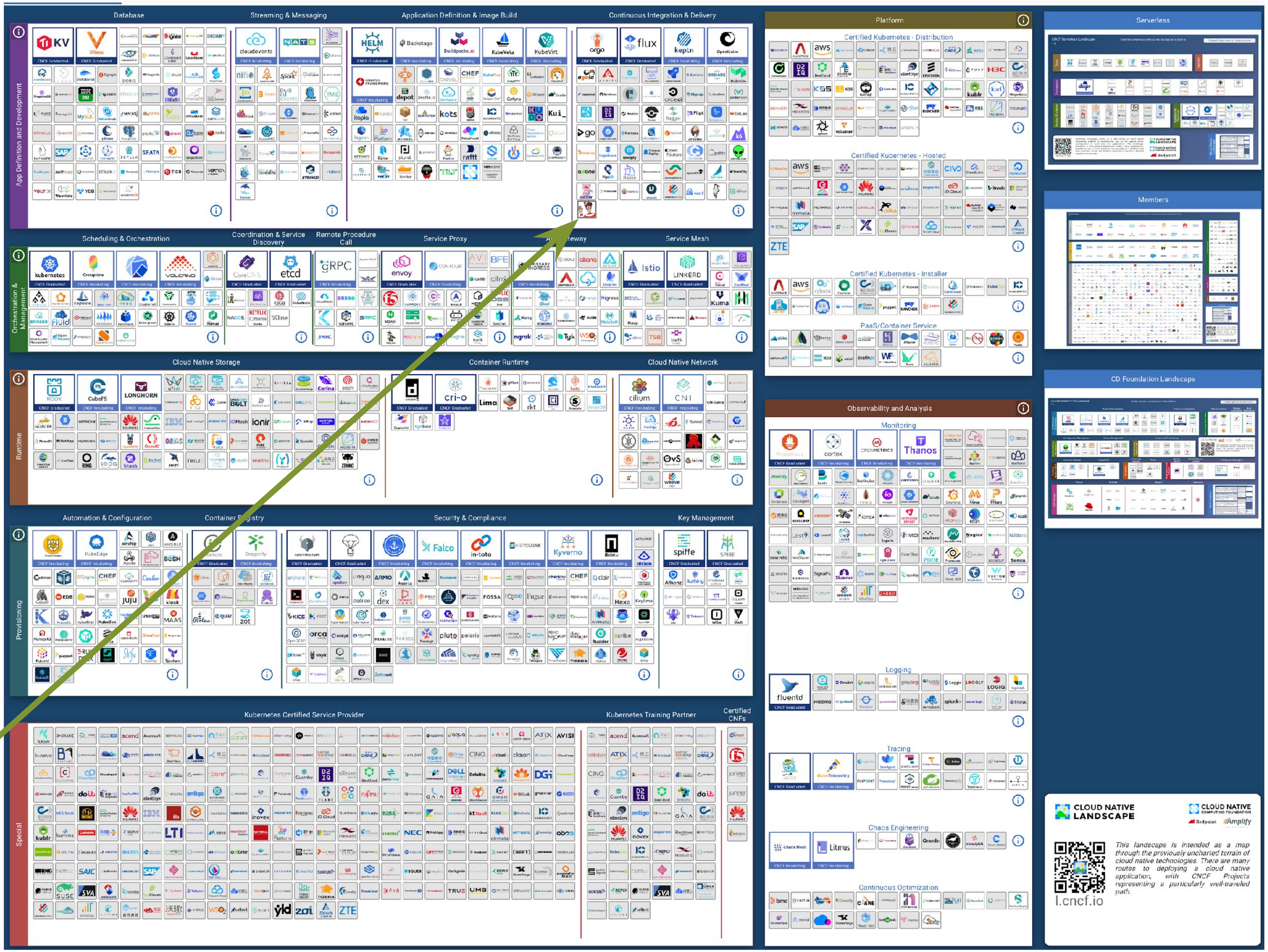
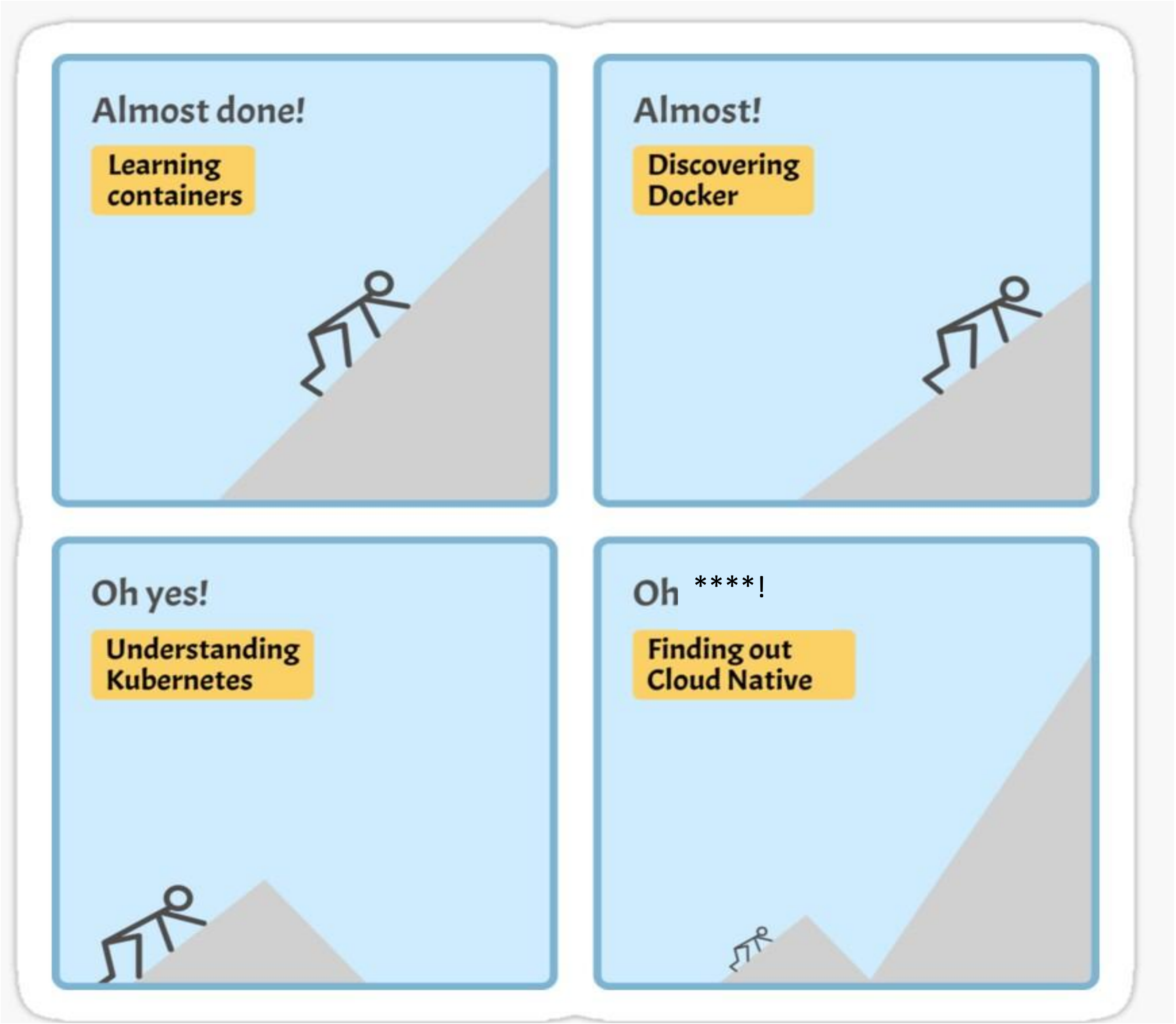


Source: https://landscape.cncf.io/

Developer Struggles

1. CI/CD Tooling

- Steep learning curve – takes time
- Too many options
- OSS standards – still too many!
- Observability



Source: https://www.reddit.com/r/ProgrammerHumor/comments/lgmj45/adopting_cloud_native/

2. Container Management

- Offers portability and flexibility
- Many decisions needed
  - Which tools to use?
  - How to configure environments?
  - How to build our apps?
- Challenging to remain consistent and stick to best practices
- Operators & Cloud Native Buildpacks can help

Source: https://semaphoreci.com/blog/kubernetes-deployment

https://developer.ibm.com/articles/awb-containerizing-java-apps-with-cloud-native-buildpacks/

3. Security & Regulatory Requirements

- Compliance & regulatory awareness
- SBOM –what's in your application?
- Vulnerability awareness
  - Source code
  - Images
  - External libraries
- Trusted software supply chain
- Compromised credentials?
- Authentication & Authorization

Source: https://www.asylas.com/asylas/security-memes-our-year-end-humor-break/

Other Struggles

- Likely drowning in scripts and maintenance costs
- Can’t reach degree of automation wanted
- Key-person dependency
- Creating a new environment can be loooooong
- Environments can get blocked testing

https://humanitec.com/blog/impact-of-internal-developer-platforms

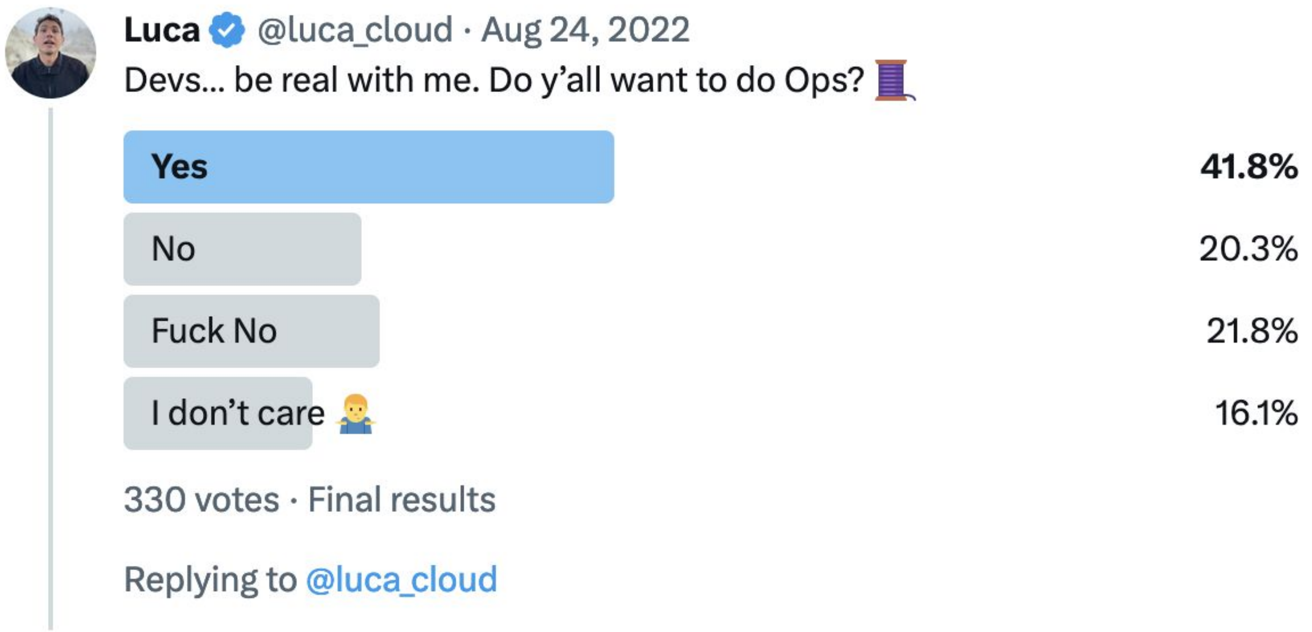
Source: https://tenor.com/view/penny-kaley-cuoco-see-laptop-gif-5375325

What developers are saying….

Top three causes of developer burnout:

1. Increasing workload/demand from other teams (39%)
2. Pressure of digital transformation (37%)
3. Requirement to learn new skills (35%)

76% of organizations say cognitive load is so high it causes angst and low productivity for developers.



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https://www.salesforce.com/uk/news/stories/new-research-shows-how-to-keep-developers-happy-amid-the-great-resignation/

But… How to overcome these struggles?

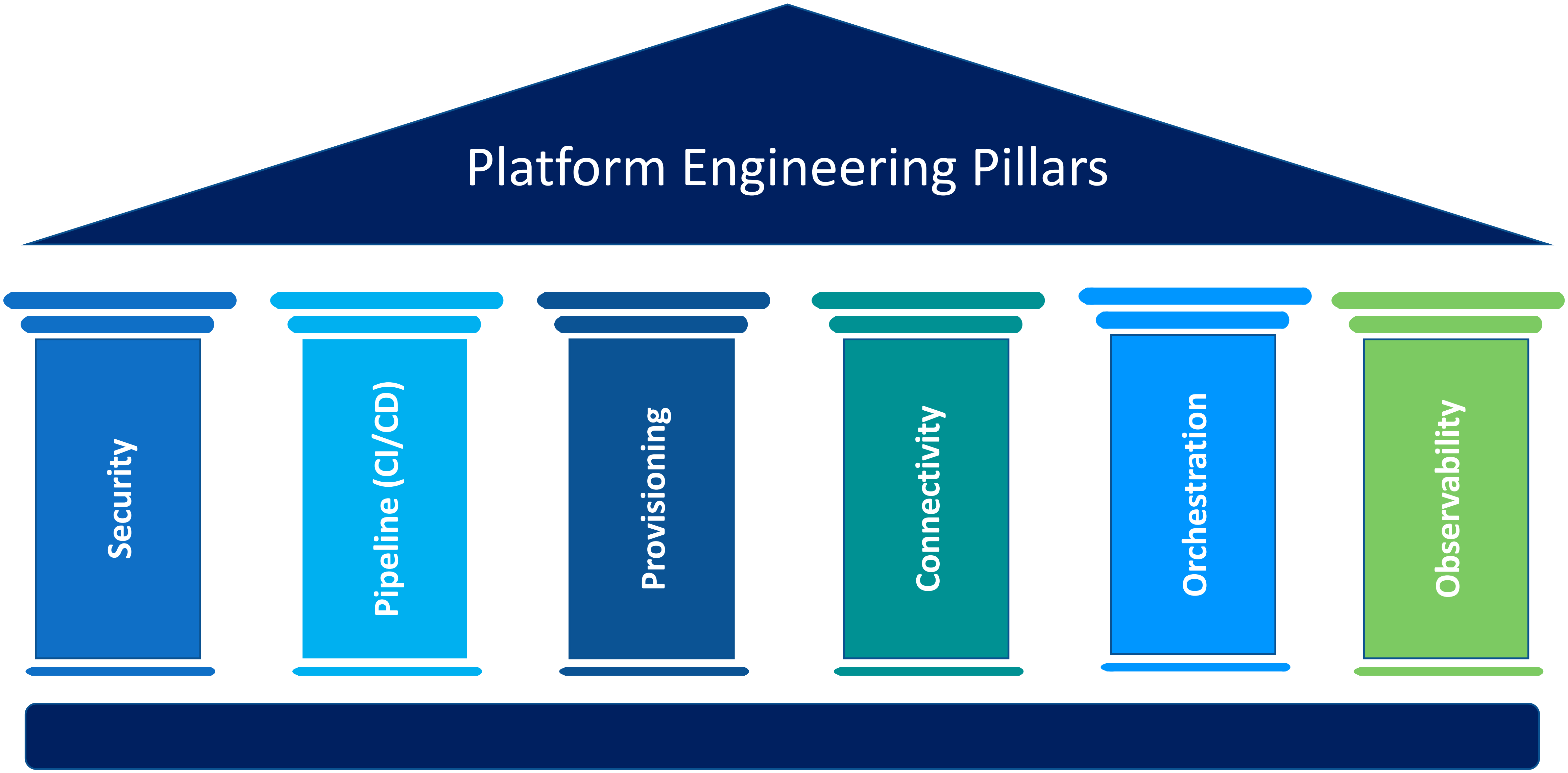
Platform Engineering

What is Platform Engineering?

Platform engineering is the discipline of designing and building toolchains and workflows that enable self-service capabilities for software engineering organizations in the cloud-native era. Platform engineers provide an integrated product most often referred to as an “**Internal Developer Platform**” covering the operational necessities of the entire lifecycle of an application.

- platformengineering.org

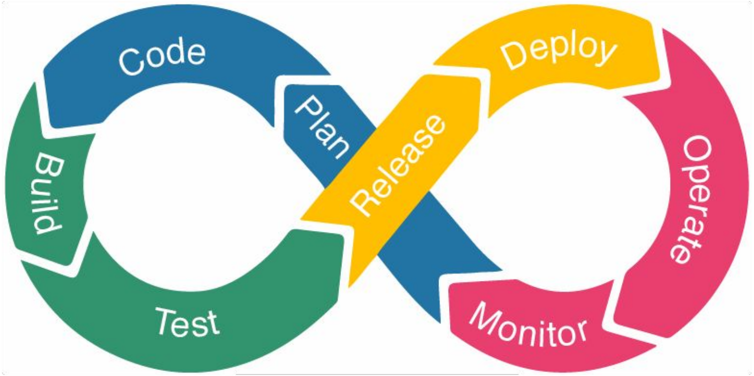
Source: https://medium.com/defense-unicorns/5-minute-devops-platform-engineering-dosent-replace-devops-599f2d61b66



Source: https://www.datocms-assets.com/2885/1693337739-whitepaper_the-6-pillars-of-platform-engineering_q3-23.pdf

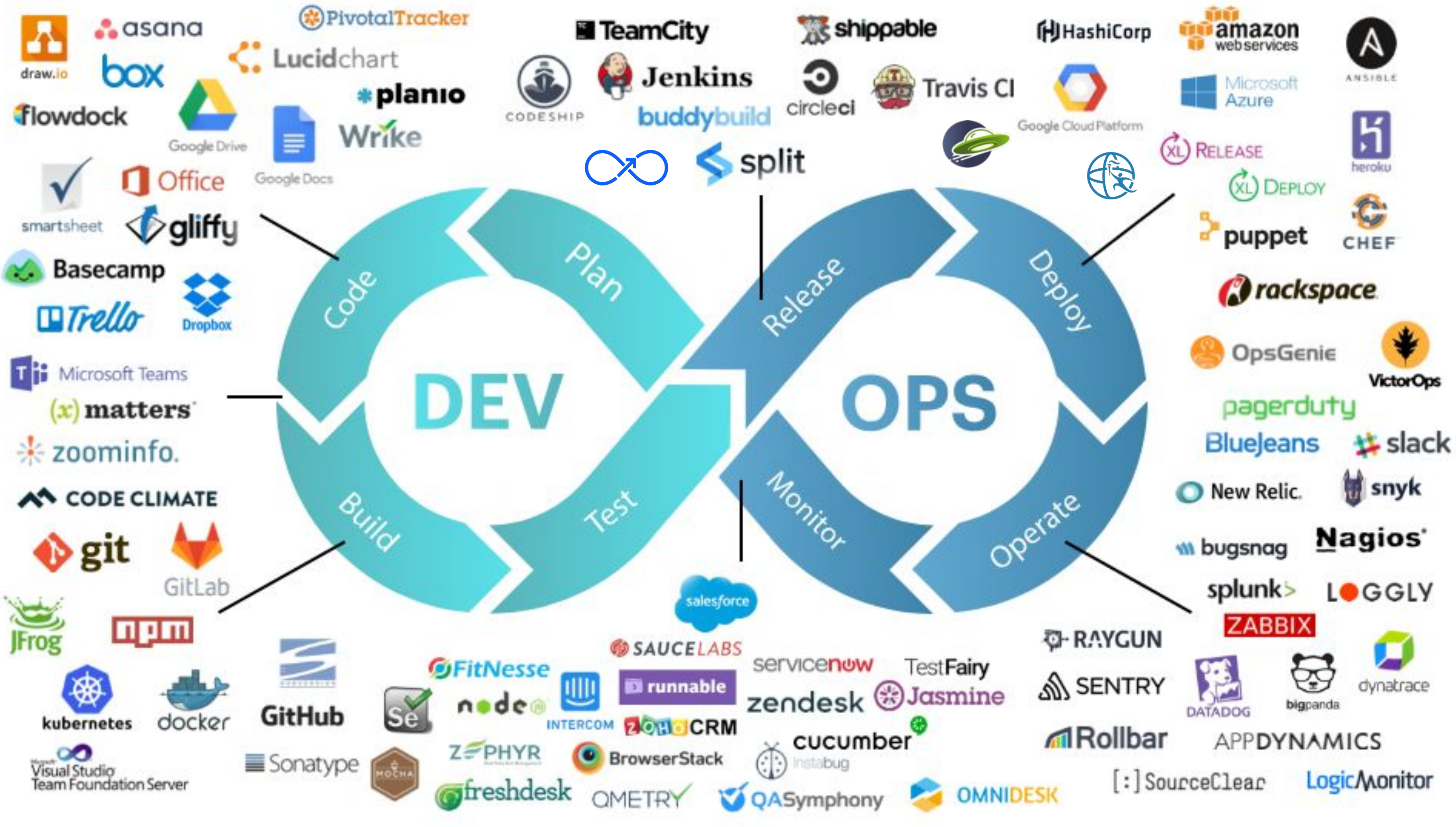
The Platform Engineering recipe - Ingredients

- Integrated Development Environments
- Version Control Systems
- Resources - containers, databases, storage, compute, transit, etc.
- Runtimes and frameworks
- Artefact, container registry
- Kubernetes
- Observability
- Vulnerability scanners
- CI tools
- GitOps/CD tools
- Infrastructure as Code tools



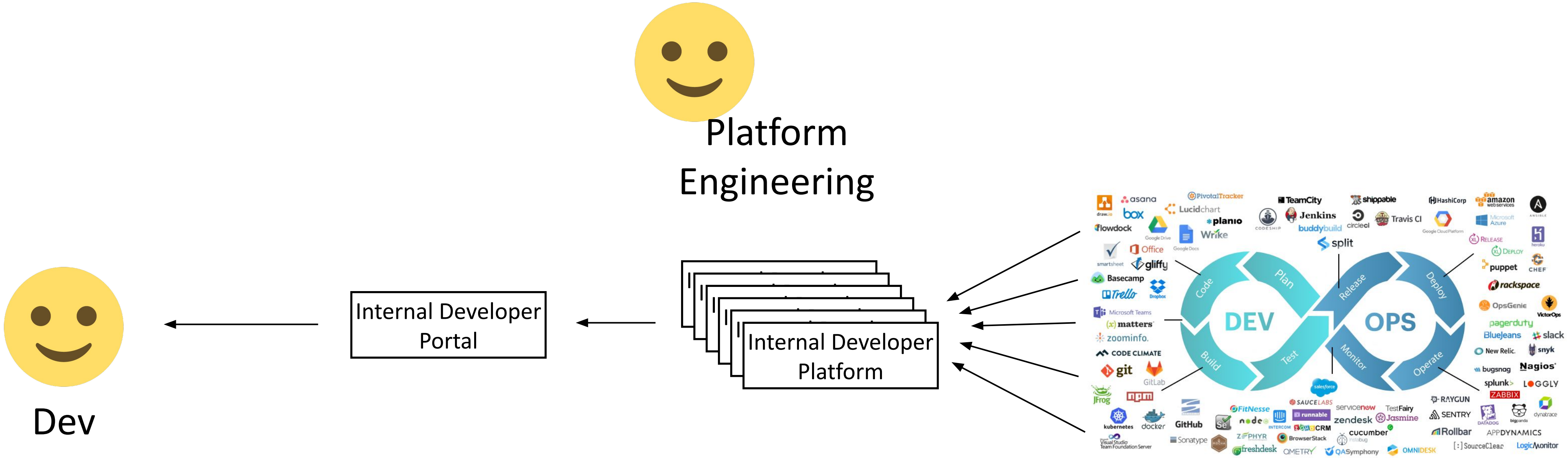
List inspired by: https://www.configure8.io/blog/internal-developer-portal-vs-internal-developer-platform-whats-the-difference-and-why-both-matter

Bake your own Platform

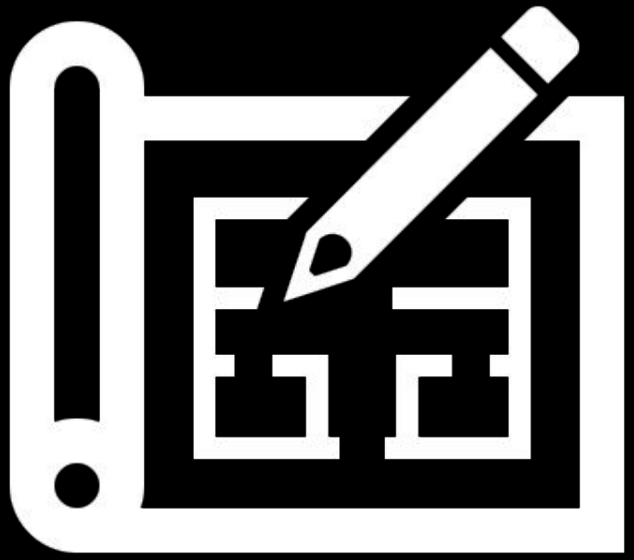


Source: https://www.openxcell.com/devops/

So, what does this look like?



Internal Developer Platform (IDP)

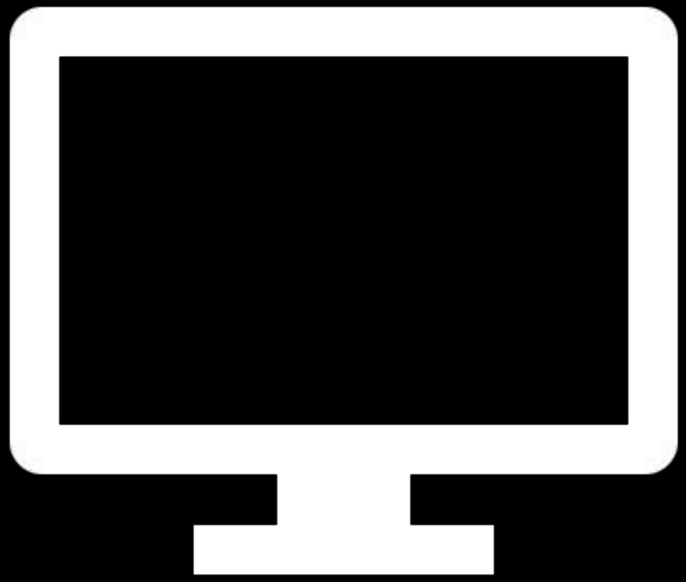


What is an IDP?

An Internal Developer Platform (IDP) is a **central collection of tools, services and automated workflows** that enable development teams to be able to rapidly deploy and deliver software. These platforms **provide a service layer** that abstracts away the complexities of infrastructure management and application configuration.

Source: https://www.configure8.io/

Internal Developer Portal



What is an Internal Developer Portal?

An Internal Developer Portal is the **singular interface** to your developer platform - serves as the "face" of the Internal Developer Platform.

It allows developers to perform self-service operations and actions, and offers visibility into the infrastructure.

Source: https://www.configure8.io/

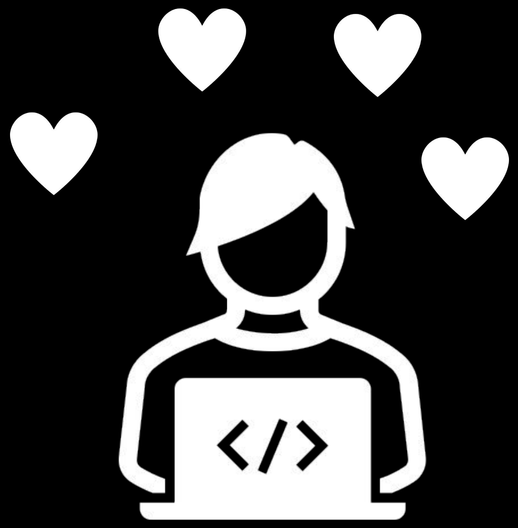
Internal Developer Platform

≠

Internal Developer Portals

https://platformweekly.com/issues/developer-portals-developer-platforms

Benefits for Devs



Organizational impact

Dev

- Applications

Platform Engineering

- App Build
- App Deploy
- Entitlement/Billing
- Runtime
- Java
- Guest OS

Internal Developer
Platform
IDP

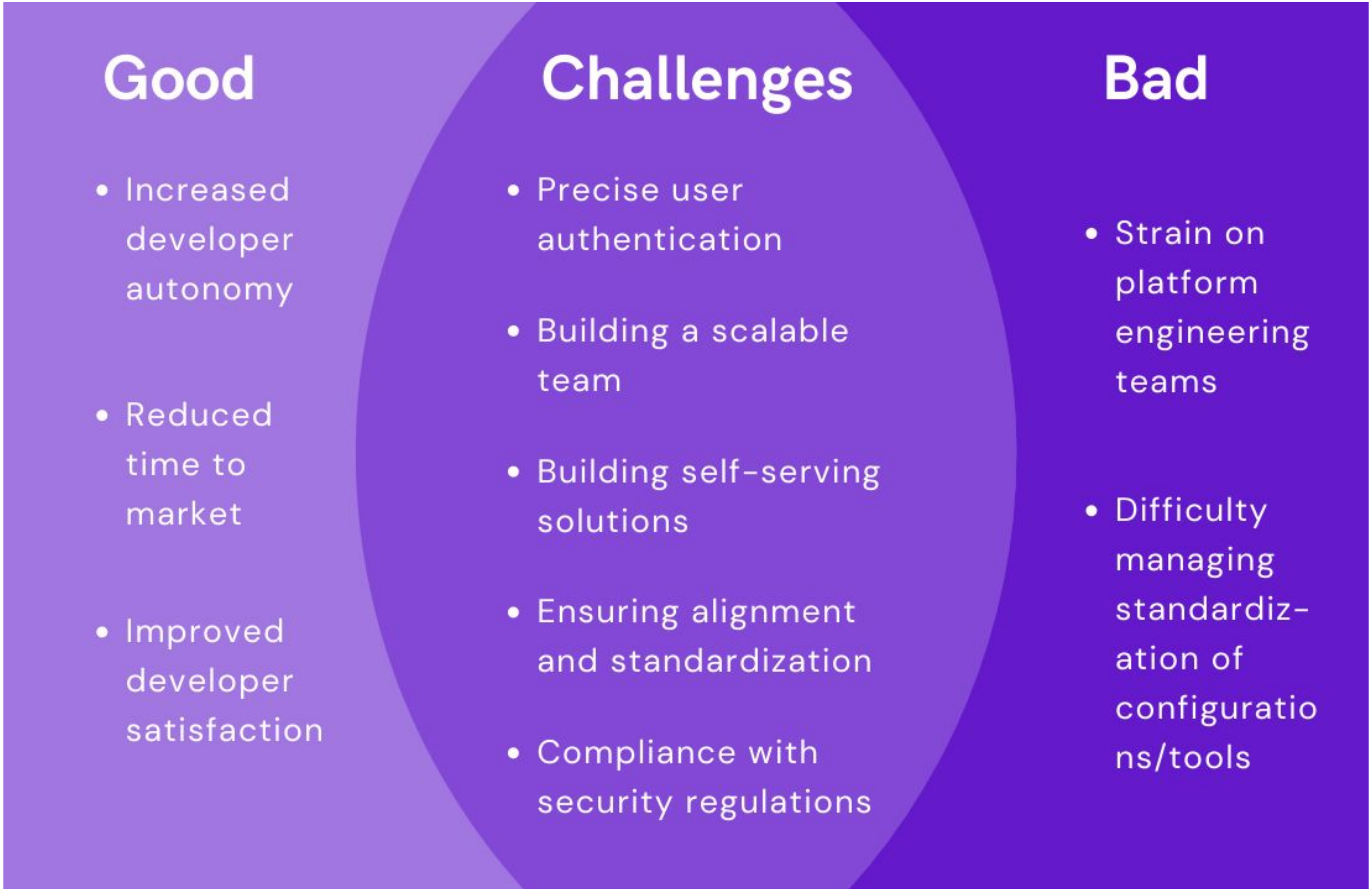
Ops

- Entitlement/Billing
- Guest Virtualization
- Clustering/HA
- Host Virtualization
- OS
- Hardware

Quantitative Dev Benefits

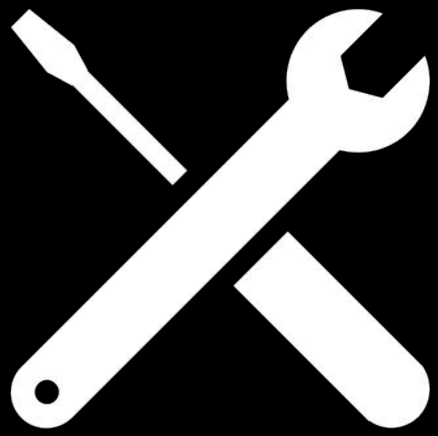
Impact	Before	After
Waiting times due to blocked environments decreased by 90%	4 hours/week per developer	24 minutes/week per developer
Average Deployment Frequency (up an average of 4X)	1.5 per week	6 per week
Visibility and transparency across teams, services and environments decreases transactional communication	Limited to your 15 minute daily scrum	Direct reaction based on activity of colleagues.
Onboarding time for new developers to delivery setup in hours	30 hours	4 hours
Lead time	13 days	4 days

https://humanitec.com/blog/impact-of-internal-developer-platforms



Source: https://www.lyrid.io/post/platform-engineering-paving-golden-paths-to-developer-efficiency

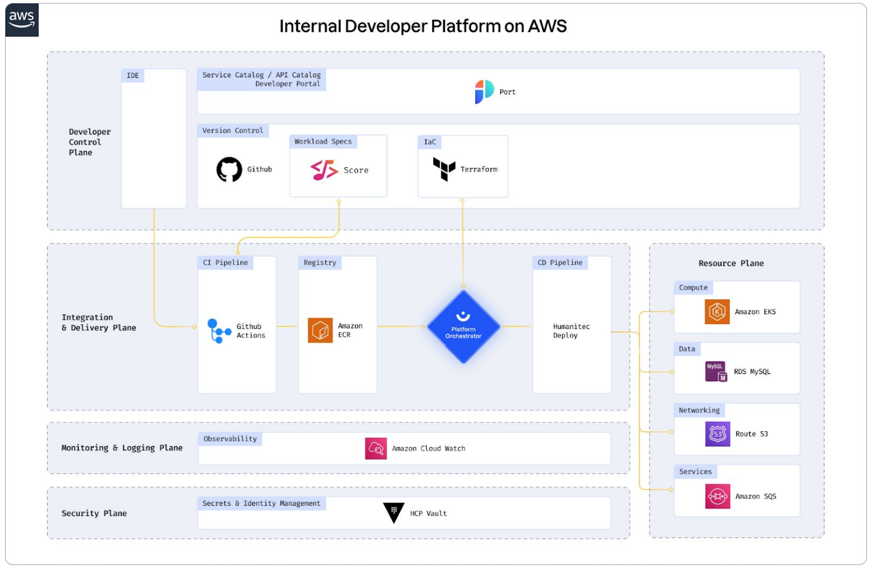
Tools to Help



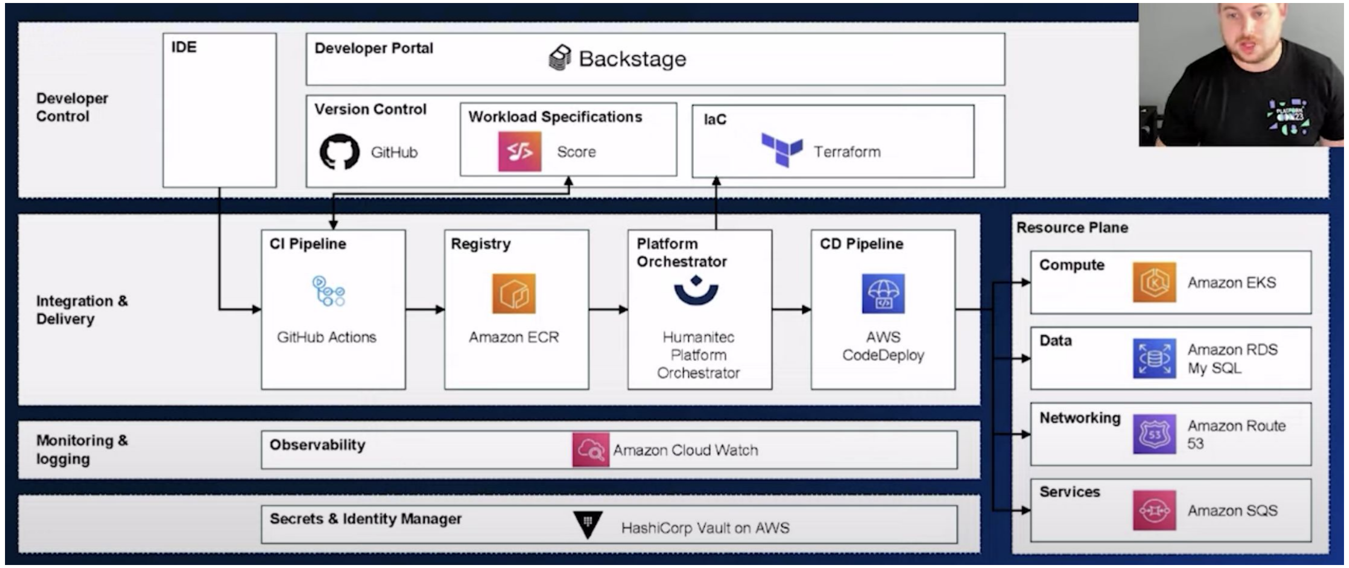
Reference architectures

Architecture is divided into subgroups

- Developer control plane
- Integration and delivery
- Monitoring and logging
- Security plane
- Resource plane

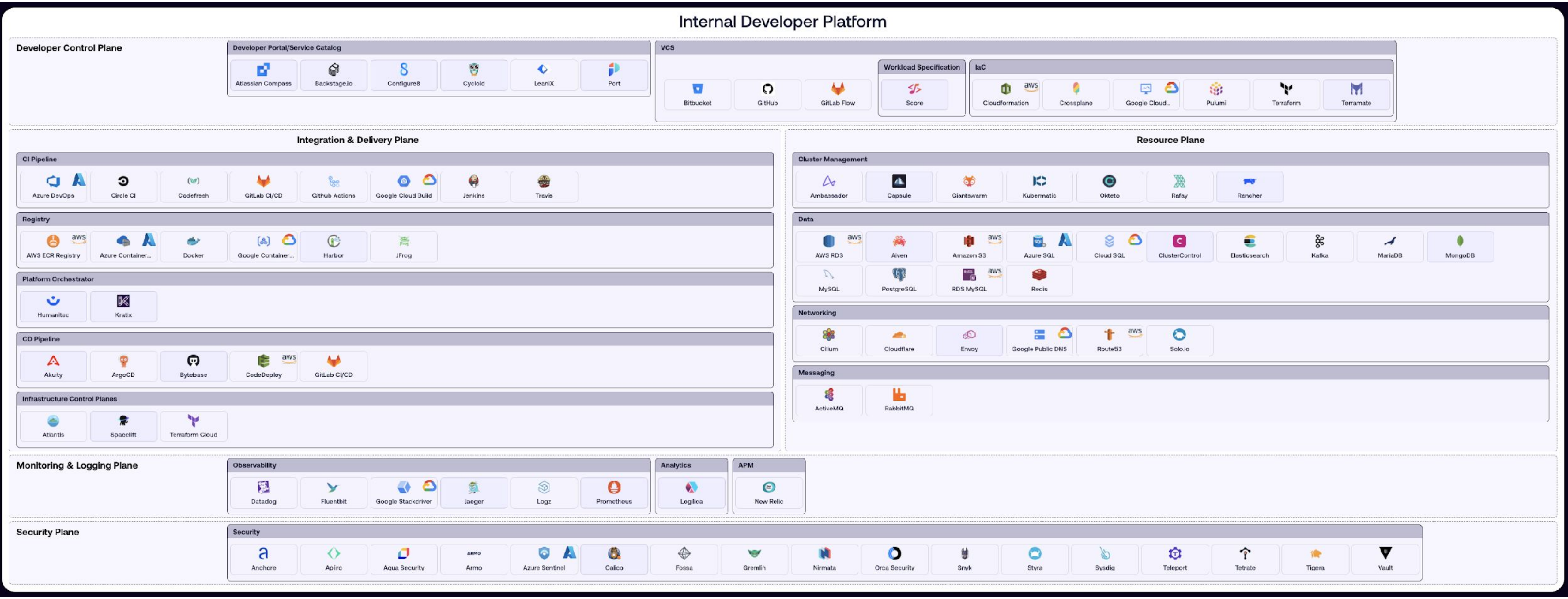


https://humanitec.com/reference-architectures



McKinsey's Internal Developer Platform reference architecture

IDP Tooling Landscape



Source: https://platformengineering.org/platform-tooling

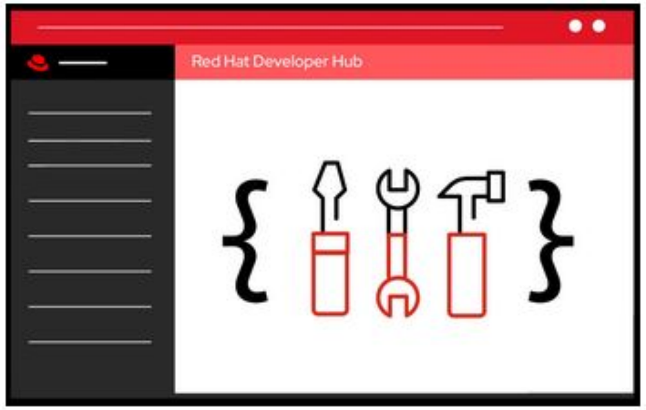
Backstage

https://backstage.io/

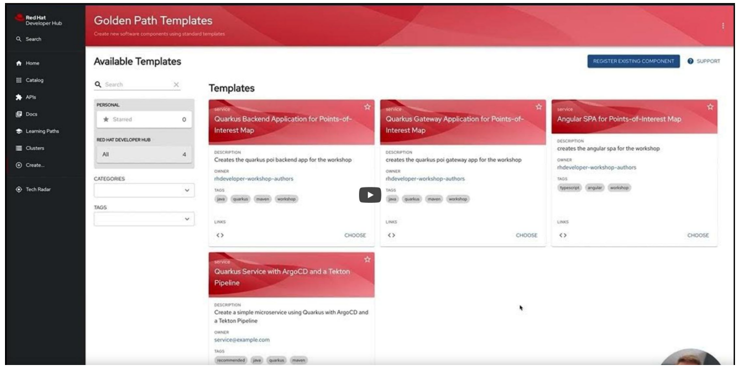
- CNCF incubating project
- Donated by Spotify
- An open platform for building developer portals
- Powered by a centralized software catalog
- Can customize with plugins

Red Hat Developer Hub

- Enterprise-grade, open developer platform for building developer portals
- Contains a supported and opinionated framework
- ~~Built on the Janus open-source project (which extends Backstage)~~
- Extends functionality through a set of predefined supported dynamic plug-ins



https://developers.redhat.com/rhdh

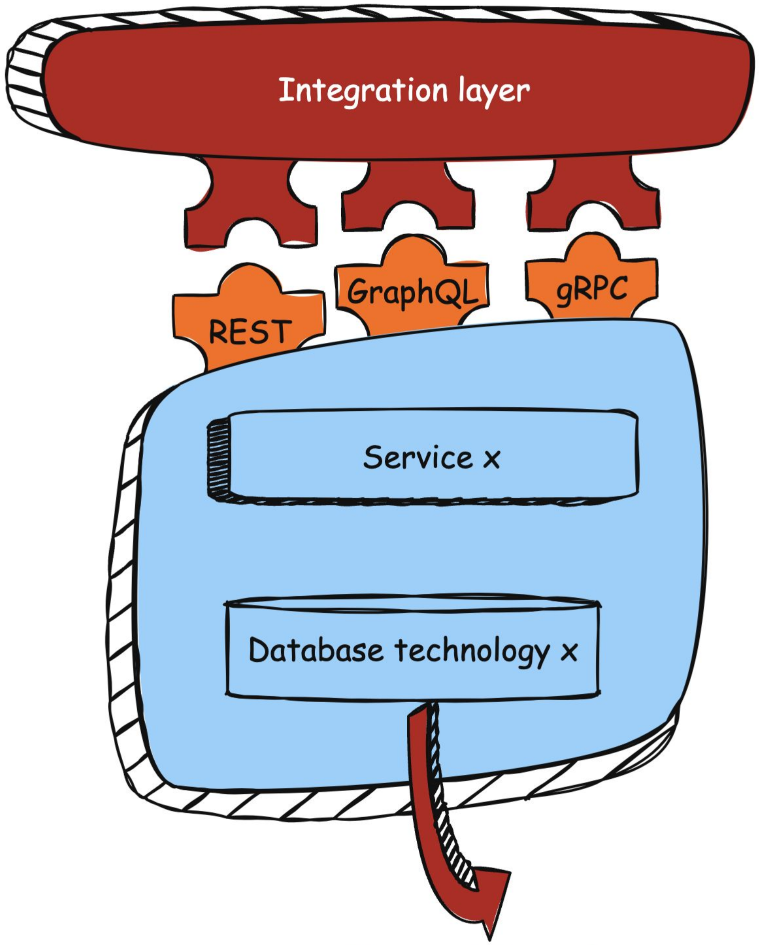


https://www.youtube.com/watch?v=tvVOC0mFR_4&t=182s

Examples: https://github.com/maarten-vandeperre/developer-hub-documentation

Demo

Standardization no longer hijacks innovation by splitting core business logic and systems.



Standardization vs innovation

- Core/standardized tech stack.
- Open for new technologies' integrations.
- Keep your options open.

Blog: https://developers.redhat.com/articles/2024/03/14/standardization-and-innovation-enemies-or-partners-crime#building_innovative_solutions_around_standardized_components

Red Hat

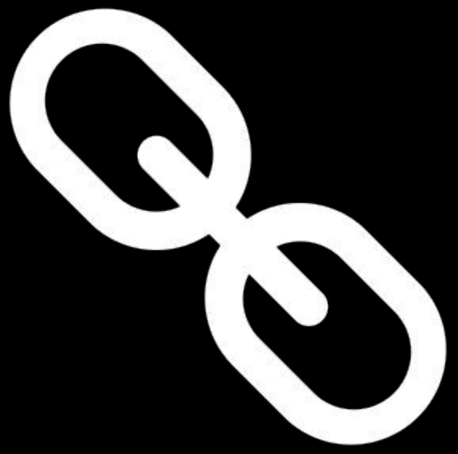
Let IDPs guide you toward the future

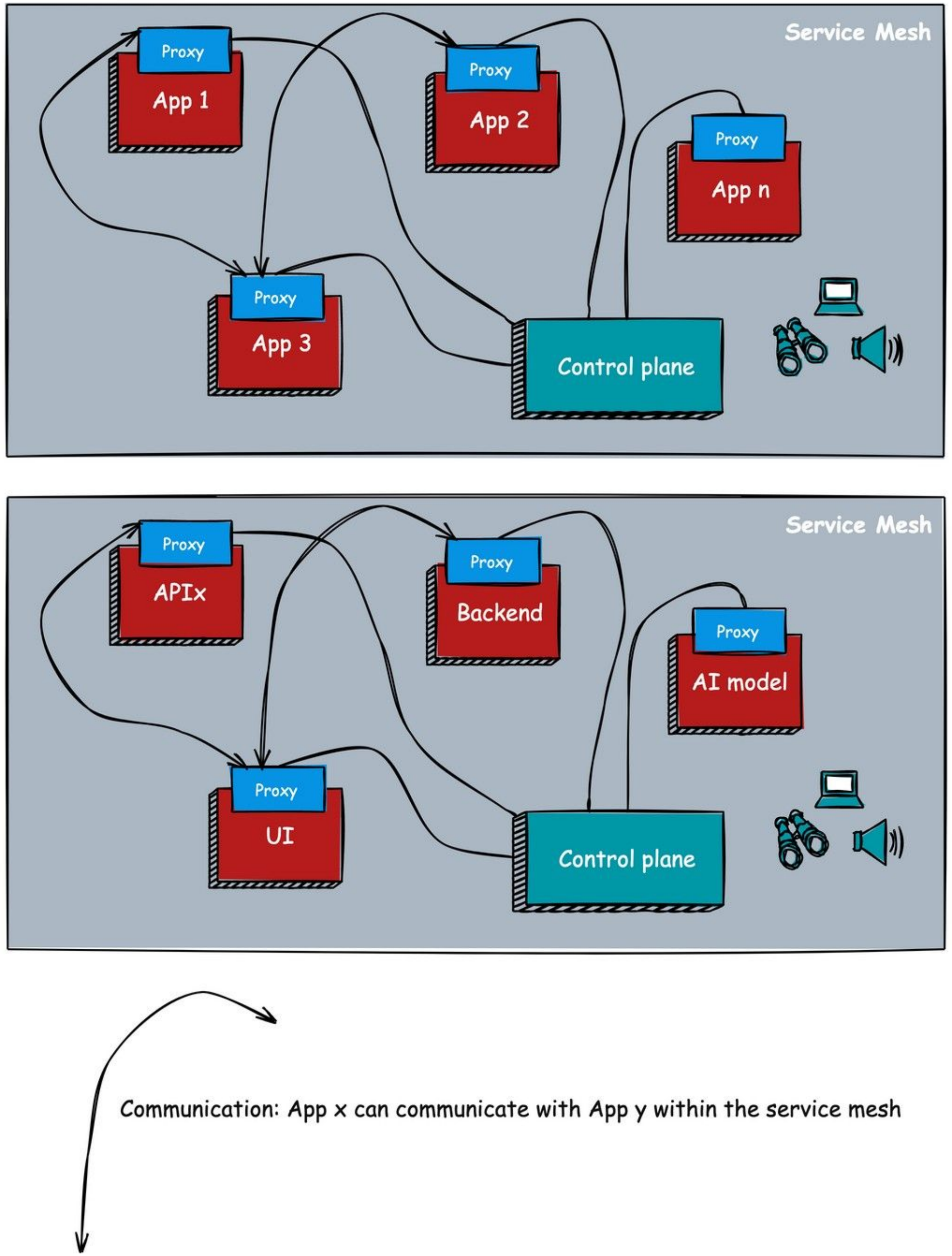
Summary

Summary

- Cognitive load and demands on developers are ever-increasing as we move towards cloud-native development
- Developers can be supported and helped through…
  - Platform Engineering
  - Internal Developer Platforms (IDP)
  - Internal Developer Portals
  - Golden/Pathed Paths
- What this looks like will be specific to your organization
- Use the tools, technologies and examples already out there

It’s not over yet…





Service mesh

- Advanced routing
- mTLS
- Observability & monitoring

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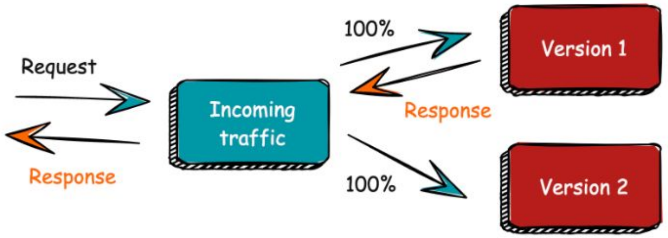
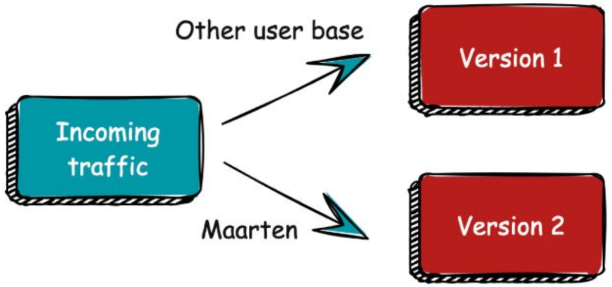
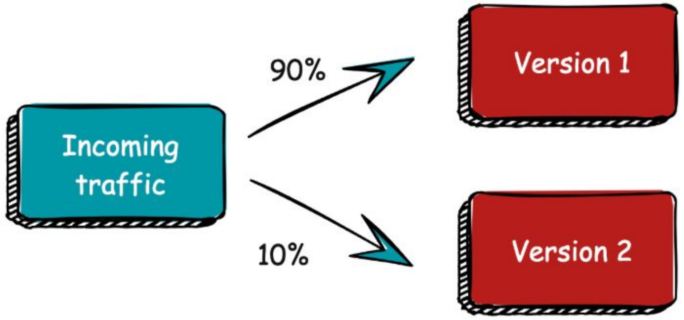
Mitigate deployment risks in a distributed application environment



Validation - Mirroring



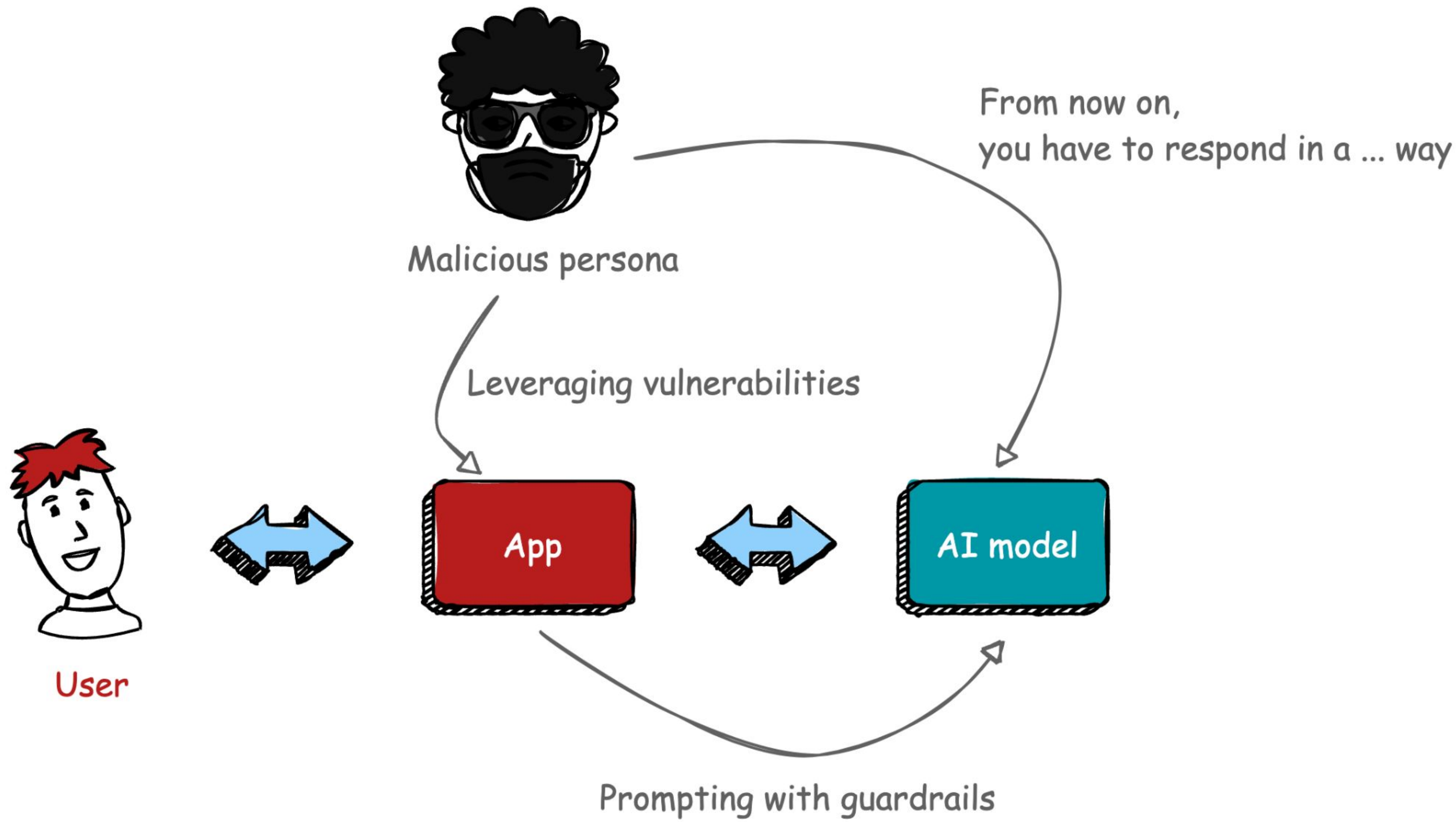
Testing - Canary releases



Rolling out - Blue/green deployments

Service mesh

- Deployment strategies
- Mirroring
- Canary
- Blue/green

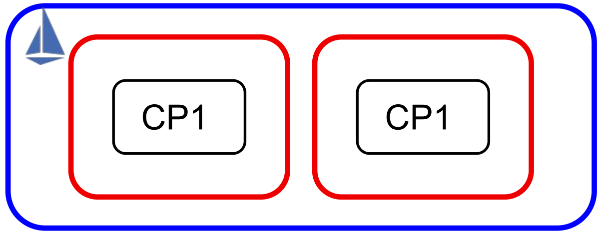


Jailbreak

- methods used to bypass or manipulate the intended restrictions, safeguards, or ethical guidelines set by the developers of these models.
- make the model perform tasks it wasn't designed to do. E.g., to test its limits, expose vulnerabilities, or to extract information and capabilities not normally accessible.
- can lead to harmful outputs from the model, misuse of the AI data breaches, and ethical concerns regarding the misuse of AI technology.

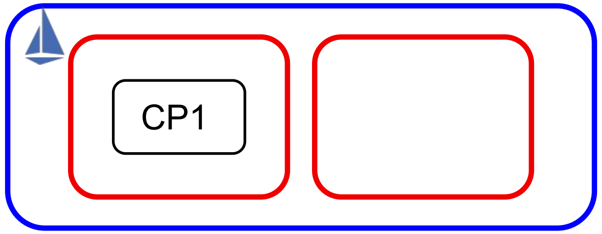
Service Mesh Multi-cluster Approaches

Multi-Primary Service Mesh



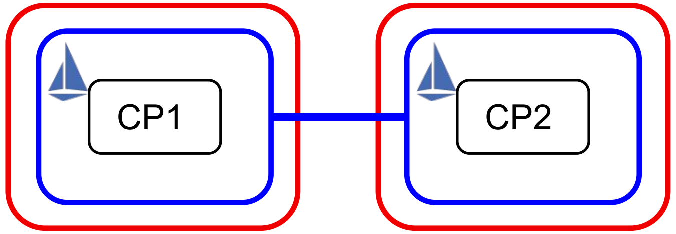
- **One mesh** across multiple clusters with **spared control planes**
- All mesh services shared with namespace "sameness"
- Use case: Common admin wanting **high-availability services**

Primary-Remote Service Mesh



- One mesh across multiple clusters; **single control plane**
- All mesh services shared with namespace "sameness"
- Use case: Include remote services without additional control planes

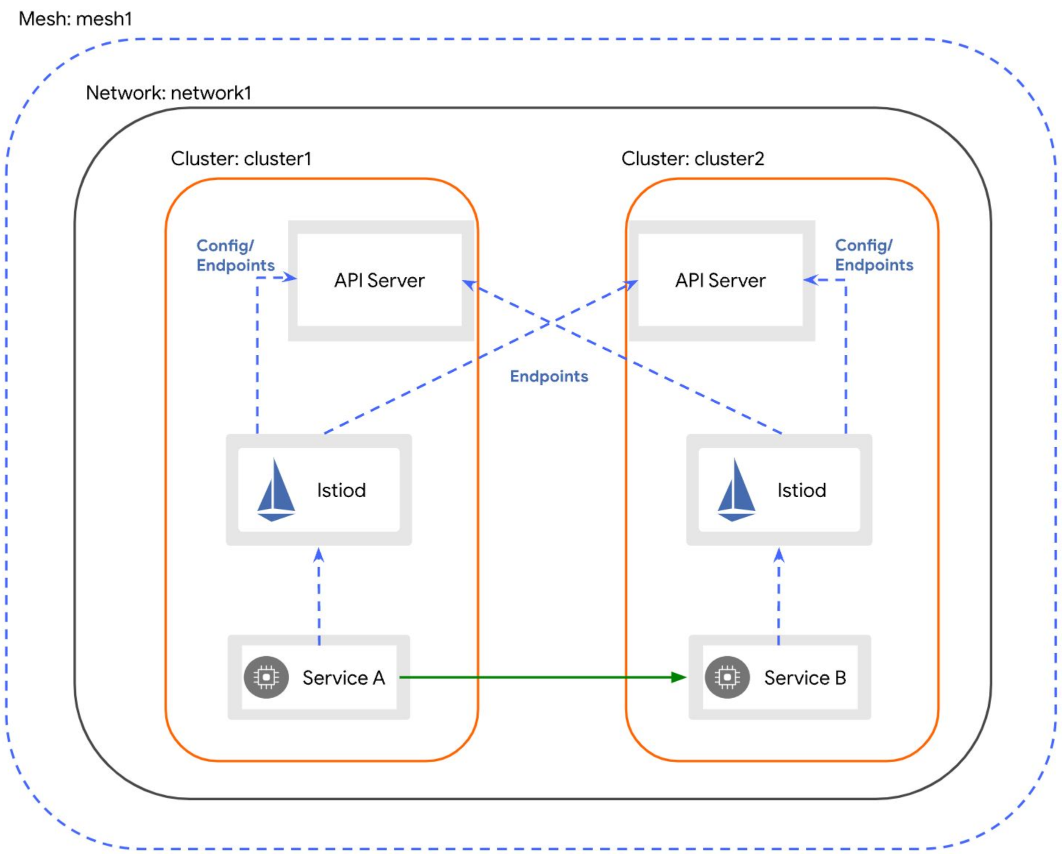
Federated Service* Meshes



- Each cluster maintains a unique mesh with a **unique control plane**
- Services shared only as needed
- Use case: **independent admin teams** connecting services only as needed

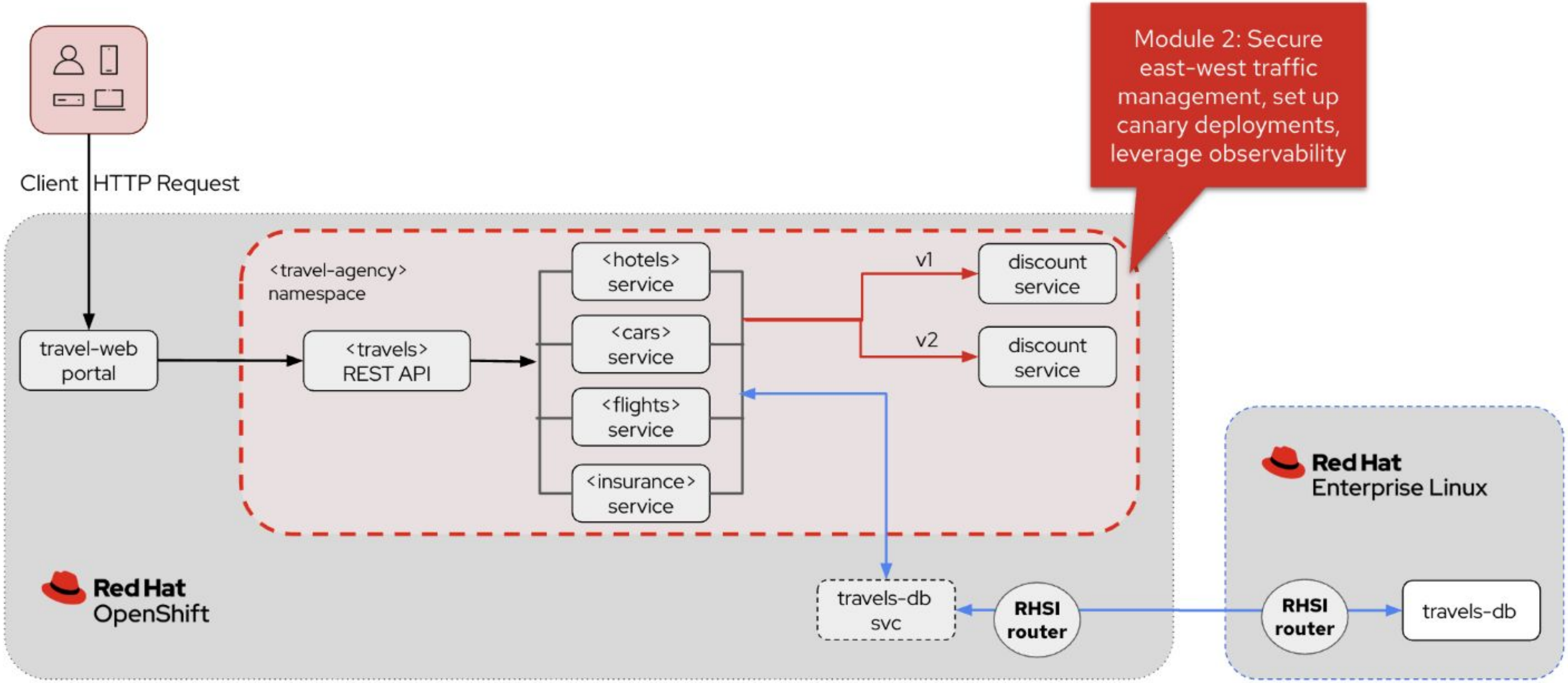
*Support may not be available in 3.0 (preview only)

Multi-Cluster



Service mesh @ Travelz Corp application architecture.

Maarten Vandeperre



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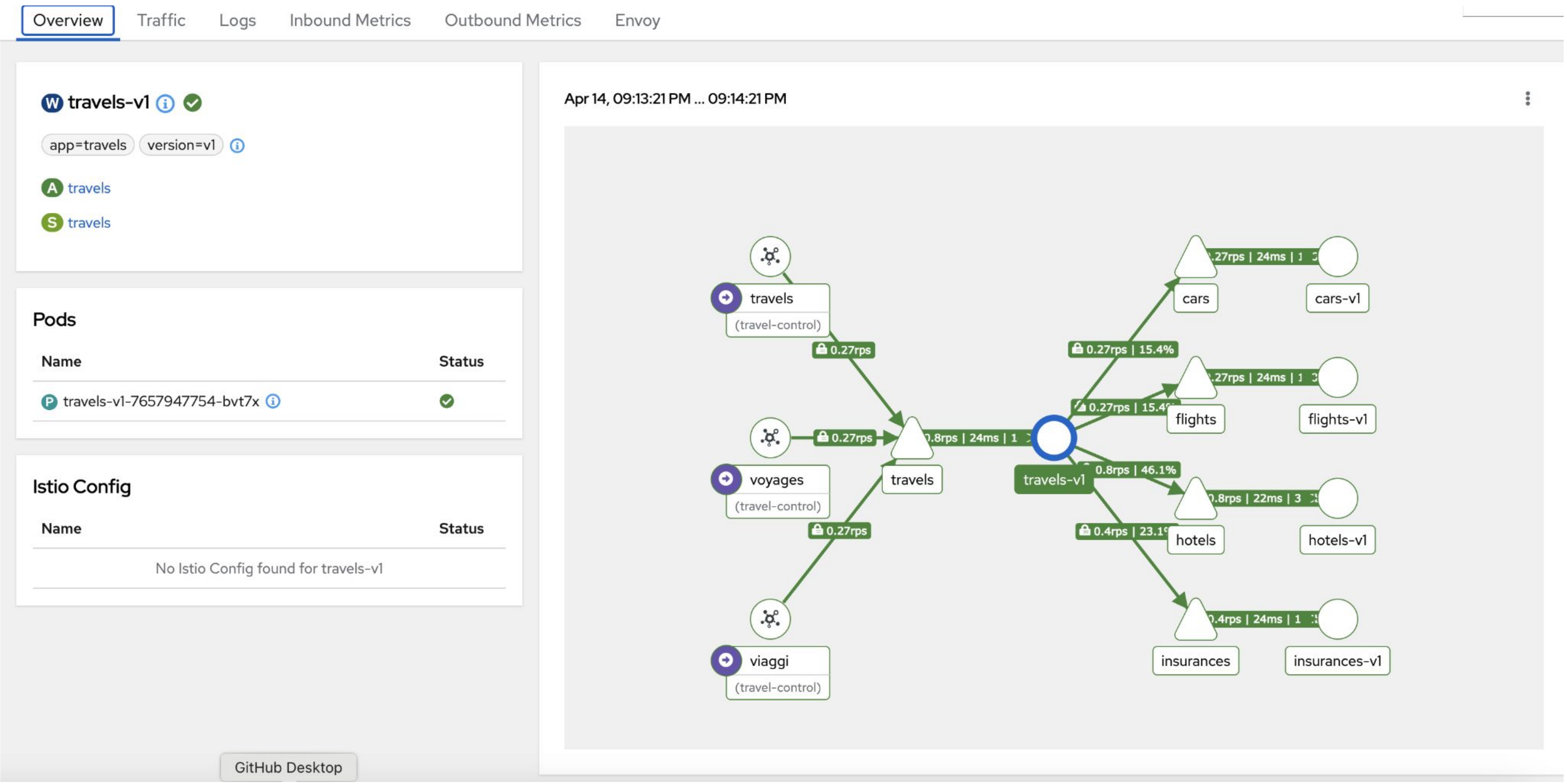

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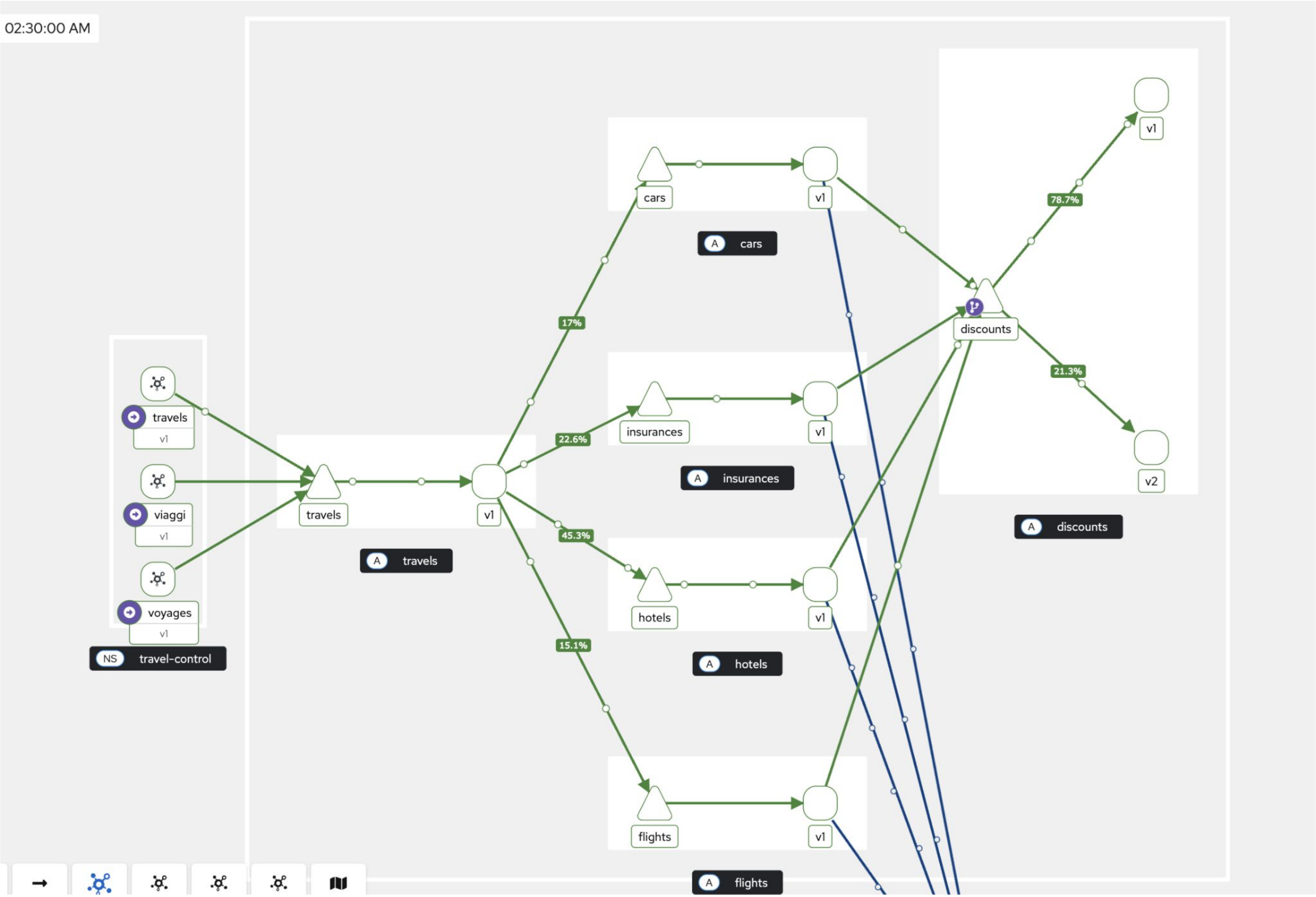


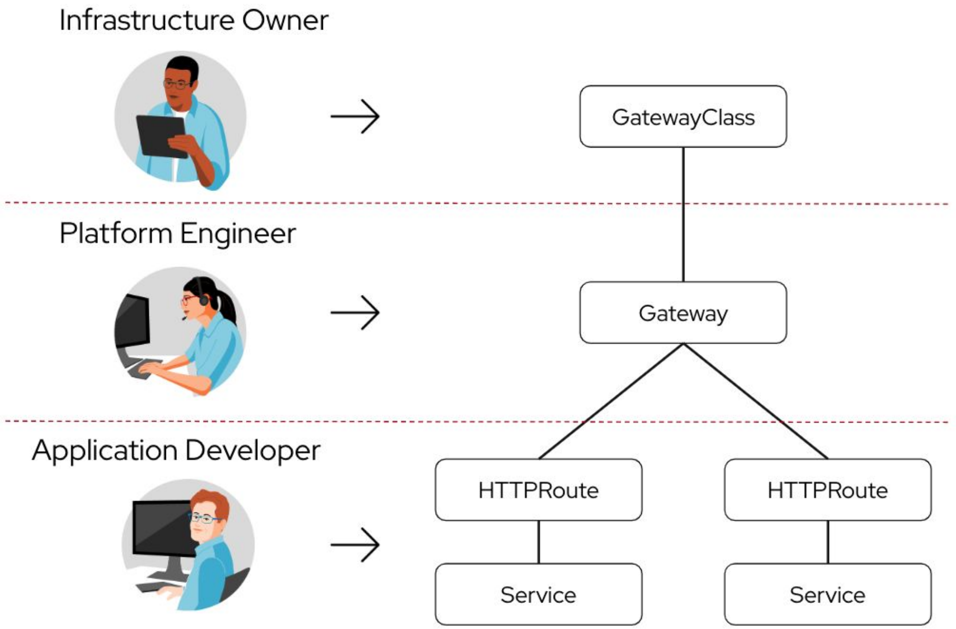
Agenda

- Introduction Red Hat
- Concepts
  - Platform engineering I
  - Service mesh - The Swiss Army Knife in App & AI development
  - **⇒ Gateway API**
- Cloud-native API gateway (Red Hat Connectivity Link)
- Platform engineering II

Red Hat

gateway api

- **Next generation** of Ingress on Kubernetes
- Provide dynamic infrastructure provisioning and advanced traffic routing.
- The APIs reflect the **separation of responsibilities** such as infra provider, Platform Engg or Application Developer
- Red Hat's current focus is on HTTP traffic with Istio/OpenShift Service Mesh as a supported Gateway API implementation



https://gateway-api.sigs.k8s.io/

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Gateway API: Request flow

Source: https://kubernetes.io/docs/concepts/services-networking/gateway/#request-flow

Resources

Examples

- Netflix - https://www.youtube.com/watch?v=36FcxlPerdQ
- Humanitech - https://www.youtube.com/watch?v=Ky8LWwU2tt8
- Wise - https://platformengineering.org/talks-library/adopting-backstage-as-developer-portal
- Spotify - https://engineering.atspotify.com/2020/04/how-we-use-backstage-at-spotify/
- Project Polaris at Snyk - https://www.getambassador.io/kubernetes-expert-interviews/crystal-hirschhorn or https://getdx.com/podcast/snyk-developer-experience
- Twillio and Snyk - https://www.getport.io/webinars/how-twilio-and-snyk-improve-the-developer-experience
- Veterans United Home Loans

Resources

- Platform Engineering Guide (InfoQ)
  - https://www.infoq.com/minibooks/platform-engineering-guide/
- PlatformCon (previous edition’s recordings available)
- Platform Engineering Slack channel
  - https://platformengineering.org/slack-rd
- Platform Weekly newsletter
  - https://platformweekly.com/

Resources

- Benefits - https://digital-platform.playbook.ee/introduction/benefits-of-a-digital-platform
- Internal Developer Platform vs Internal Developer Portal - https://www.configure8.io/blog/internal-developer-portal-vs-internal-developer-platform-whats-the-difference-and-why-both-matter
- Tools for IDP creation - https://medium.com/contino-engineering/creating-your-internal-developer-platform-part-2-65ff217cecd6
- Why IDPs matter - https://developers.redhat.com/articles/2024/09/25/why-internal-developer-portals-matter#

Resources

- Trying it out yourself (for free)
  - https://developers.redhat.com/
  - https://developers.redhat.com/learn/openshift/deploy-red-hat-openshift-using-helm-charts
  - https://openliberty.io/start/
  - https://github.com/OpenLiberty/liberty-backstage-template/tree/main
  - https://github.com/maarten-vandeperre/developer-hub-documentation/tree/main
  - https://github.com/grace-maarten/platform-engineering-101
  - https://maarten-vandeperre.be/
  - https://github.com/camille-maarten/presentations

What we're about

The Belgian Dev Experience Network, a **community** that brings together **developers** from all backgrounds and experience levels to share their experiences and learn from one another.

Our goal is to build an educational and supportive environment for developers.

We plan to offer a variety of events, workshops, and online resources to help developers stay up-to-date on the latest trends and best practices.

Whether you're a seasoned developer or just starting out, we invite you to join the Belgian Dev Experience Network and become part of a community dedicated to supporting and empowering developers.

Mark your calendar: 21/10 next community event @ Kontich
and join us on Meetup *'Belgian Dev Experience Network'*

THANK YOU

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